



UNITY UNIVERSITY

DATABASE SYSTEMS INDIVIDUAL ASSIGNMENT

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Multi-user DBMS Architecture

A Multi-User Database Management System architecture refers to the structure or design of a database system that allows multiple users to access and manipulate data concurrently. There are common architectures that are used to implement multi-user data base management system: for example, teleprocessing, file-server, database server, and client-server.

1 Client-Server: In a multi-user DBMS architecture, the system typically follows a client-server model. There are one or more database servers that store and manage the data, and multiple clients that connect to these servers to perform operations on the data. The client-server is a computing architecture that divides tasks or processes between clients and servers. In this model, clients request services or resources from servers, which provide those services or resources in response to client requests. This client-server model can give us many advantages

- It enables wider access to existing databases
- It increased performances
- Increased consistency

2 Teleprocessing: it refers to the use of telecommunication networks to enable remote access to a centralized database system. Teleprocessing in DBMS architecture allows users at remote locations to interact with the database system, perform transactions, and retrieve information without physically accessing the central database server.

3 File-server: a file server is a component that provides access to shared files and data stored on a networked file system. While a file server itself is not a traditional part of a DBMS, it can play a role in certain architectures, particularly in scenarios where a more traditional file-based approach is used alongside or instead of a full-fledged database system. In a file-server the processing is distributed about the network mainly in a local area network. It holds the files required by the applications and the DBMS run on each workstation. The file server model have many advantages for example

- File-server architecture is relatively simple to set up and maintain compared to more complex database management systems (DBMS).
- There will be a large amount of network traffic The file-server architecture mode provides a straightforward and cost-effective solution for storing, accessing, and sharing files within a networked environment, making it suitable for small to medium-sized businesses, workgroups, and departments within larger organizations.

4 Database server: the database server is a central component responsible for storing, managing, and providing access to the database. It serves as the backend infrastructure that handles all interactions with the database, including data storage, retrieval, manipulation, and security enforcement. Database server have many advantages, for example;

- **Centralized Data Management** A database server provides a centralized platform for storing, managing, and accessing data. This centralization simplifies data management tasks by consolidating data in one location, making it easier to maintain, secure, and backup.
- **Data Integrity** Database servers enforce data integrity constraints defined in the database schema, ensuring that the data remains consistent and accurate over time. This includes enforcing entity integrity, referential integrity, and other data validation rules to prevent data corruption and maintain data quality.
- **Concurrency Control** Database servers implement concurrency control mechanisms to manage concurrent access to the database by multiple users or client applications. This ensures that transactions execute in a consistent and isolated manner, preventing data conflicts and maintaining data integrity.

In addition the use of a multi-user DBMS architecture provides different benefits and uses including concurrent access, improved data sharing, data consistency, centralized data management, scalability, performance optimization, enhanced security, support for complex applications, and data backup and recovery capabilities. These advantages make multi-user DBMS architectures well-suited for organizations of all sizes and industries. Let us see some uses of multi-user of **DBMS** architecture.

- **Concurrent Access:** Multiple users can access and manipulate the database simultaneously. This allows for collaborative work environments where teams can work on shared data sets concurrently without conflicts.
- **Improved Data Sharing:** Multi-user DBMS architectures facilitate data sharing among users within an organization. Users can access and share data across departments, teams, or geographical locations, promoting collaboration and information exchange
- **Data Consistency:** Multi-user DBMS architectures enforce data consistency by managing concurrent access to the database. Mechanisms such as locking, transactions, and isolation levels ensure that data remains consistent and accurate, even when accessed by multiple users simultaneously.